MIL-E-1/240C(ER)
17 May 1983
SUPERSEDING
MIL-E-1/240B(EL)
20 November 1970

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6098 1/

This specification is approved for use by the Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Amplifier, beam power

Outline --- See figure 1

Base --- B8-58, modified (low-loss phenolic)

Envelope --- T11

Cathode --- Coated unipotential

Base connections:

Pin No.	1	3	5	6	7	8
Element	k,g3	a	g2	h	g1	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec2	Ec1	Ic1	Pp	Pg2	Ehk	Ik	TE	epy	Alt
Unit:	V	V dc	V dc	V dc	mA dc	W	W	v	mA dc	°C	v	ft
Maximum:												
Class A:	6.9	400	315	---	0	21	3.5	200	125	225	---	See note 1
Class B:	6.9	600	315	---	---	21	3.5	200	60	225	1,250	See note 1
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	250	250	-22.5	---	---	---	---	---	---	---	---

GENERAL:

Preproduction sample approval - Required (see note 2)

1/ Formerly 6098/6AR6WA

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Insp level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1								
1301	Heater current			0.65	II	If	1.08	1.32	A
1336	Heater-cathode leakage			0.65	II	Ihk	---	75	µA dc
1266	Total grid current (1)	3		0.65	II	Ic1	---	-1.0	µA dc
1256	Electrode current (1) (anode)			0.65	II	Ib	55	100	mA dc
1256	Electrode current (screen)			0.65	II	Ic2	---	8.0	mA dc
---	High voltage	4	Ec1 = -150 V dc; Ec2 = 300 V dc; Ebb = 1,250 V dc; Ic1 = 0.2 mA dc; R1 = 5,000 ohms	0.65	II	e	1,150	---	v
1306	Transconductance (1)			0.65	II	Sm	4,300	6,500	µmhos
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes			2.5	I	R	25	---	Meg
1256	Electrode current (2) (anode)		Ec1 = -65 V dc	2.5	I	Ib	---	1.0	mA dc
1266	Total grid current (2)	5	Ef = 7.0 V	2.5	I	Ic1	0	-3.0	µA dc
1306	Transconductance (2)		Ef = 5.7 V	2.5	I	ΔSm Ef	---	15	%
1246	Audio frequency noise		Ec1 = -35 V dc; Rp = 2,000 ohms; Ecal = 245 mV ac	2.5	I	EB	---	17	vu
1031	Low-frequency vibration		Ec1 = -35 V dc; Rp = 2,000 ohms	6.5	Code H	Ep	---	1,000	mV ac
1331	Direct-interelectrode capacitance		No shield	6.5	Code E	Cgp	0.4	0.9	pF
			No shield			Cin	8.5	13.5	pF
			No shield			Cout	5.0	9.2	pF
---	Ripple	6	Eb = 325 V dc; Ec2 = 300 V dc; Rp = 1,000 ohms; Rk = 500 ohms; Ef = 6.3 V ac, 400 Hz	6.5	Code H	Ep	---	100	mV ac

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Insp level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 - Continued								
1031	Vibration-fatigue			6.5	See note 7	---	---	---	---
1041	Shock		450 G; Ehk = 100 V dc; Rg1 = 0.1 Meg	6.5	See note 7	---	---	---	---
---	Vibration-fatigue and shock-test end points								
1031	Low-frequency vibration		Ec1 = -35 V dc; Rp = 2,000 ohms	---	---	Ep	---	1,500	mV ac
1336	Heater-cathode leakage			---	---	Ihk	---	150	µA dc
1266	Total grid current (1)			---	---	Ic1	0	-2.0	µA dc
1306	Transconductance (1)			---	---	Sm	3,500	---	µmhos
1101	Secureness of base, cap, or insert			---	---	---	---	---	---
1111	Base pin solder depth			---	---	---	---	---	---
1105	Permanence of marking			---	---	---	---	---	---

Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1516	Stability life		Ehk = 200 V dc; Rg1 = 47,000 ohms; TA = room	---	---	---	---
---	Stability life-test end point						
1306	Transconductance (1)			ΔSm_t	---	10	%
1501	Intermittent life	8,9	Ehk = 200 V dc; Rg1 = 47,000 ohms; TE = 225°C (min)	---	---	---	---

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Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3 - Continued						
---	Intermittent life-test end points (500 hours)	10					
1301	Heater current			If	1.08	1.32	A
1336	Heater-cathode leakage			Ihk	---	75	µA dc
1266	Total grid current (1)			Ic1	0	-2.0	µA dc
1306	Transconductance (1)			Sm	4,000	6,500	µmhos
1306	Transconductance (1), average change			Avg ΔSm t	---	15	%
1211	Insulation of electrodes			R	10	---	Meg
1306	Transconductance (2)		Ef = 5.7 V	ΔSm Ef	---	15	%
---	Intermittent life-test end points (1,000 hours)	11					
1301	Heater current			If	1.08	1.32	A
1336	Heater-cathode leakage			Ihk	---	75	µA dc
1266	Total grid current (1)			Ic1	0	-3.0	µA dc
1306	Transconductance (1)			Sm	3,700	6,500	µmhos
1031	Variable-frequency vibration	12	Ec1 = -35 V dc; Rp = 2,000 ohms	Ep	---	1,000	mV ac
1216	Base material insulating quality	12	Low-loss phenolic, zone 11 (min)	---	---	---	---

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. Preproduction or first article sample inspection shall conform to the requirements of appendix F of MIL-E-1 and shall consist of performing all tests specified in this TSS. Invitation for bids should provide that the preparing activity reserves the right to waive the requirements for preproduction or first article samples as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending acquisition.
3. This test shall be performed at the conclusion of the holding period.
4. This test shall be made before the transconductance (1) and transconductance (2) tests. With a 60-Hz sinusoidal driving voltage applied to the control grid, measure the peak to peak voltage across the load.
5. Prior to this test, tubes shall be preheated 5 minutes at conditions indicated below. Test within 3 seconds after preheating.

Ef V ac	Ec1 V dc	Rk Ohms	Rg1 Meg	Eb V dc	Ec2 V dc
7.0	-22	0	0.1	250	250

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6. Adjust Ec1 so that Ib = 55 mA dc. Measure ripple between anode and ground with meter having an impedance of 100,000 ohms or greater. The heater shall be tied to negative end of cathode resistor.

7. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

9. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples and shall not exceed a total of three tube failures.

10. The allowable defectives per characteristic shall be not greater than one for the first sample and shall be not greater than three for combined samples.

11. The allowable defectives per characteristic shall be not greater than two for the first sample and shall be not greater than five for combined samples.

12. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, AQL 6.5, inspection level S3. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

13. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - ER

Review activities:
Army - AR, MI
DLA - ES

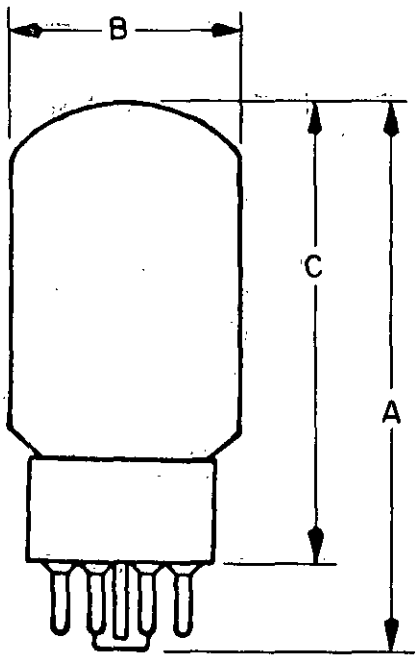
User activity:
Army - AL

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5960-A187)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		3.50 (88.90)
B		1.44 (36.58)
C		2.94 (74.68)

FIGURE 1. Outline drawing of electron tube type 6098.

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